SWAPNALI NAGESH PATKI

Seattle, WA, USA | snpatki1@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Data Scientist/Machine Learning Engineer experienced in operationalizing enterprise-grade LLM and CV systems, applied causal inference, econometrics-informed statistics, delivering Revenue Operations and Business Analytics through Responsible AI.

PROFESSIONAL EXPERIENCE

Data Scientist / Machine Learning Engineer | ***Western EcoSystems Technology Inc, Seattle, WA*** *Jan 2023 - Jan 2026*

- Generated $250k+ in new revenue and secured 4 enterprise clients by partnering with senior leadership to establish AI roadmaps.
- Engineered and productionized 10+ multi-modal computer vision systems for classification, object detection, segmentation and multi-object tracking in Azure ML, evaluated with k-fold cross-validation, A/B testing, and MLflow, improving F1 score by 20-25%.
- Fine-tuned and deployed an end-to-end GenAI RAG pipeline using Python, Pinecone and GitLab CI/CD to transform lengthy, technical RFPs into concise summaries for business stakeholders, eliminating 400+ hours of manual analysis.
- Modeled and integrated an end-to-end synthetic data generation pipeline using Stable Diffusion to realistically simulate rare wildlife age and sex classes, boosting minority class F1-score by 20% while reducing classification model drift and bias.
- Reduced manual field survey time by 60% by building and deploying a computer vision carcass-detection model at solar facility, processing GIS data (ArcGIS Pro, ArcPy, Python, R) with field technicians, GIS, and drone teams to support regulatory compliance.
- Accelerated deployment speed by 2x, optimizing CLIP-based transformer inference for semantic segmentation with CUDA-memcheck that supported statistical analysis on sustained regulatory compliance across wind-turbine facilities.
- Led a KPI-driven data quality and cross-functional automation framework across data engineering and annotation teams, identified operational bottlenecks and reducing manual annotation effort by 45%.
- Presented customer-facing business analytics using SQL, Python, Tableau, and Power BI to translate data into actionable insights.

Machine Learning Researcher | ***Stevens Institute of Technology, AISec Lab, Hoboken, NJ*** *Jun 2022 - Dec 2022*

- Achieved suicide-risk sentiment detection F1 scores from 0.78 to 0.85 across 1M+ social media posts by designing an end-to-end data pipeline on Apache Flink with feature engineering and ensembled forecasting models.
- Quantified cascading failure patterns across 10,779 transmission outages and 6,687 cascading events using Python, NetworkX, and spatial propagation analysis, visualizing results in Gephi/Tableau and presenting findings at a university research symposium.

Software Engineer | ***L&T Technology Services, Mumbai, India*** *Jan 2021 - Aug 2021*

- Supported 100K+ active users under high-throughput production workloads, as measured by system stability and response-time SLAs, by engineering backend systems in Java and optimizing API execution pipelines, SQL query performance and orchestration.
- Improved company website load time by 30% and increased capacity to handle higher traffic volumes by designing a MySQL database for optimized data retrieval, building full-stack features in .NET, Angular, and React, validating APIs with Postman.
- Cut incident resolution time 40% by automating production monitoring and failure detection workflows using Splunk and JIRA.

Software Development Engineer | ***Persistent Systems, Pune, India*** *Jun 2019 - Dec 2019*

- Prevented recurring reporting discrepancies of up to $25,000 per batch by identifying and resolving a data mismatch between PostgreSQL tables and Power BI dashboards used for business reporting.
- Engineered backend microservices and REST APIs for an optimized search engine using Node.js, TypeScript, C#, and MySQL, improving query response time by 20% through redesigning indexing logic, optimizing database retrieval patterns.

EDUCATION

Westcliff University, Irvine, CA | Master of Business Administration, Applied AI in Business**, GPA: 4.0** *Jan 2026 - present*
Stevens Institute of Technology, Hoboken, NJ | Master of Science, Applied Artificial Intelligence**, GPA: 3.9** *Aug 2021 - Dec 2022*
Savitribai Phule Pune University, India | Bachelor of Engineering, Information Technology**, GPA: 7.8** *Aug 2016 - May 2020*

PROJECTS

ETF Portfolio Optimizer Using LLM with Time-Series Analysis and Price Monitoring

- Fine-tuned FinGPT using LoRA adapters on 500 ETF funds and macroeconomic data via RAG for real-time recommendations.
- Developed 5-15 minute CI/CD price-monitoring pipeline with TimescaleDB and yfinance to support dynamic rebalancing decisions.

TECHNICAL SKILLS

Languages: Python, R, SQL (Snowflake, MySQL, PostgreSQL, NoSQL), PyTorch, TensorFlow, Scala, C++, C#, Java, JavaScript, TypeScript
Data Science and Machine Learning: Hugging Face Transformers, Retrieval-Augmented Generation (RAG), vLLM, CLIP, LangChain, Llama, NLP, Vector Embeddings, YOLO, SAM, Bayesian methods, Time Series (ARIMA), Statistical Inference, A/B Testing
Developer Tools: Apache Spark, Kafka, Hadoop, Snowflake, Voxel51, Docker, Kubernetes, MLflow, Kubeflow, Jenkins, GitLab CI/CD, CircleCI, AWS (SageMaker, EKS, Lambda, S3, EC2), GCP (Vertex AI, BigQuery), Azure ML, Tableau, Power BI, JetBrains Junie